

LAPORAN PRATIKUM
PRAKTIKUM ELEKTRONIKA DIGITAL

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No. _____
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Biner ke desimal

1. 101 = $1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 = 4 + 0 + 1 = 5$
2. 1001 = $1 \times 2^3 + 0 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 = 8 + 0 + 0 + 1 = 9$
3. 1101 = $1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 = 8 + 4 + 0 + 1 = 13$
4. 1010 = $1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 = 8 + 0 + 2 + 0 = 10$

Hexa ke biner (menggunakan tabel Hexadesimal ke biner,
ppt dari bu novri, eldig teori)

B4A = 101101001010
3A1 = 001110100001
E24 = 11100100100
F16 = 111100010110

Okta ke biner

68 = ?

$$712 = \left| \begin{array}{c} 111 \\ 7 \end{array} \right| \left| \begin{array}{c} 001 \\ 1 \end{array} \right| \left| \begin{array}{c} 010 \\ 2 \end{array} \right| = 111001010$$

$$212 = \left| \begin{array}{c} 010 \\ 2 \end{array} \right| \left| \begin{array}{c} 001 \\ 1 \end{array} \right| \left| \begin{array}{c} 010 \\ 2 \end{array} \right| = 010001010$$

$$25 = \left| \begin{array}{c} 011 \\ 3 \end{array} \right| \left| \begin{array}{c} 101 \\ 5 \end{array} \right| = 011101$$

Oktal ke desimal

$$134 = 1 \times 8^2 + 3 \times 8^1 + 4 \times 8^0 = 64 + 24 + 4 = 92$$

$$301 = 3 \times 8^2 + 0 \times 8^1 + 1 \times 8^0 = 192 + 0 + 1 = 193$$

$$252 = 2 \times 8^2 + 5 \times 8^1 + 2 \times 8^0 = 128 + 40 + 2 = 170$$

$$404 = 4 \times 8^2 + 0 \times 8^1 + 4 \times 8^0 = 256 + 0 + 4 = 260$$

Desimal ke biner

8

$$\begin{array}{r} 8 \\ \underline{2} 2 = 0 \\ 4 \\ \underline{2} 2 = 0 \\ 2 \\ \underline{2} 2 = 0 \\ 1 \end{array}$$

$$= 1000$$

128

$$\begin{array}{r} 128 \\ \underline{2} 2 = 0 \\ 64 \\ \underline{2} 2 = 0 \\ 32 \\ \underline{2} 2 = 0 \\ 16 \\ \underline{2} 2 = 0 \\ 8 \\ \underline{2} 2 = 0 \\ 4 \\ \underline{2} 2 = 0 \\ 2 \\ \underline{2} 2 = 0 \\ 1 \end{array}$$

24

$$\begin{array}{r} 24 \\ \underline{2} 2 = 0 \\ 12 \\ \underline{2} 2 = 0 \\ 6 \\ \underline{2} 2 = 0 \\ 3 \\ \underline{2} 2 = 1 \\ 1 \end{array}$$

$$= 11000$$

$$= 10000000$$

53

$$\begin{array}{r} 53 \\ \underline{2} 2 = 1 \\ 26 \\ \underline{2} 2 = 0 \\ 13 \\ \underline{2} 2 = 1 \\ 6 \\ \underline{2} 2 = 0 \\ 3 \\ \underline{2} 2 = 1 \\ 1 \end{array}$$

$$= 110101$$